

JIRA Story: Weather Forecast API Integration for Travel App

In the Travel Application, the Weather Forecast API feature will allow users to view current weather conditions and a 7-day forecast for their selected travel destinations. This feature enhances user experience by providing real-time weather information, helping travelers plan activities based on expected weather conditions. Below is the detailed JIRA story for implementing this API feature.

Story Details

Title: Weather Forecast API Integration for Travel App

Story ID: TRAVELAPP-303

Assignee: To Be Assigned

Priority: High

Sprint: Sprint 10

Description

As a user,

I want to see the current weather and 7-day forecast for my selected destination,

So that I can plan my travel activities accordingly.

Acceptance Criteria

1. ****API Endpoint & Method****:

- Endpoint: `/api/v1/weather`
- HTTP Method: GET
- Description: Fetches current weather and a 7-day forecast for a specified location.

2. **Request Parameters**:

- ``location`` (string, required): City and country (e.g., 'Paris, France').
- ``unit`` (string, optional): Temperature unit, either 'metric' (Celsius) or 'imperial' (Fahrenheit).

3. **Sample Request**:

```
``GET /api/v1/weather?location=Paris,France&unit=metric``
```

4. **Response Structure**:

- **200 OK**: Successful retrieval of weather data.

```
...
```

```
{
```

```
  'current': {
```

```
    'temperature': 15,
```

```
    'condition': 'Cloudy',
```

```
    'humidity': 78,
```

```
    'wind_speed': 10
```

```
  },
```

```
  'forecast': [
```

```
    {'day': 'Monday', 'high': 18, 'low': 10, 'condition': 'Sunny'},
```

```
    {'day': 'Tuesday', 'high': 17, 'low': 12, 'condition': 'Rainy'}
```

```
  ]
```

```
}
```

```
...
```

- **400 Bad Request**: Invalid or missing location parameter.
- **500 Internal Server Error**: Server issues or API provider errors.

Additional Testing Notes

- Ensure API key is securely stored and not exposed in the client application.
- Test edge cases like invalid locations, large response times, and unsupported characters in location names.
- Use tools like Postman to validate request/response structure, HTTP status codes, and error handling.
- Optimize API call frequency to manage load, especially for frequently accessed locations.
- Implement and verify caching mechanisms for repeated queries within a short timeframe.